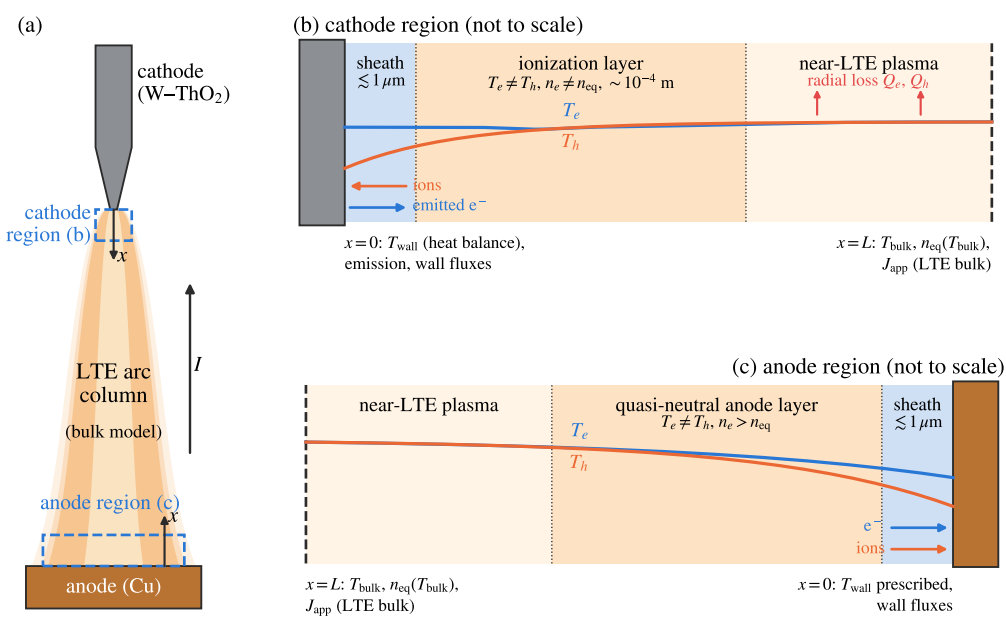
(a)
cathode
(W–ThO2)
cathode
region (b)
x
LTE arc
column
(bulk model)
I
anode region (c)
x
anode (Cu)
(b) cathode region (not to scale)
sheath
≲ 1 μm
ionization layer
Te ≠ Th, ne ≠ neq, ~ 10−4 m
near-LTE plasma
radial loss Qe, Qh
Te
Th
ions
emitted e−
x = 0: Twall (heat balance),
emission, wall fluxes
x = L: Tbulk, neq(Tbulk),
Japp (LTE bulk)
(c) anode region (not to scale)
near-LTE plasma
quasi-neutral anode layer
Te ≠ Th, ne > neq
sheath
≲ 1 μm
Te
Th
e−
ions
x = L: Tbulk, neq(Tbulk),
Japp (LTE bulk)
x = 0: Twall prescribed,
wall fluxes